Class 10 Math - Chapter 2: Polynomials

🌟 Introduction to Polynomials

A **polynomial** is an algebraic expression consisting of variables and coefficients, that involves only the operations of addition, subtraction, multiplication, and non-negative integer exponents of variables.

Key Terms:

- **Degree of a Polynomial:** The highest power of x in a polynomial $p(x)$ is called the degree of the polynomial.
- **Linear Polynomial:** A polynomial of degree 1 (e.g., $2x - 3$).
- **Quadratic Polynomial:** A polynomial of degree 2 (e.g., $x^2 + 5x + 6$).
- **Cubic Polynomial:** A polynomial of degree 3 (e.g., $x^3 - 4x^2 + x - 1$).

📌 Geometric Meaning of the Zeroes of a Polynomial

The zeroes of a polynomial $p(x)$ are precisely the x-coordinates of the points where the graph of $y = p(x)$ intersects the x-axis.

> [!NOTE] A polynomial of degree n can have at most n zeroes. For example, a quadratic polynomial has at most 2 zeroes, and a linear polynomial has exactly 1 zero.

Mind Map of Polynomial Types

```
graph TD;
    Polynomials-->Linear;
    Polynomials-->Quadratic;
    Polynomials-->Cubic;
    Linear-->Degree_1["Degree 1\n(1 Root)"];
    Quadratic-->Degree_2["Degree 2\n(Max 2 Roots)"];
    Cubic-->Degree_3["Degree 3\n(Max 3 Roots)"];
```

⚡ Relationship Between Zeroes and Coefficients

For a quadratic polynomial $ax^2 + bx + c$, if its zeroes are α and β:

1. **Sum of Zeroes:** $$ \alpha + \beta = -\frac{b}{a} $$
2. **Product of Zeroes:** $$ \alpha\beta = \frac{c}{a} $$

> [!IMPORTANT] If you are given the sum (S) and product (P) of zeroes, you can form the quadratic polynomial using the formula: $$ p(x) = k(x^2 - Sx + P) $$ where k is a constant.

Example Problem

Find the zeroes of $x^2 + 7x + 10$ and verify the relationship. **Solution:** Factorizing: $x^2 + 5x + 2x + 10 \Rightarrow x(x+5) + 2(x+5) \Rightarrow (x+2)(x+5)$ Zeroes are $\alpha = -2$ and $\beta = -5$. Sum = $-2 + (-5) = -7$. Formula: $-b/a = -7/1 = -7$. (Verified) Product = $(-2)(-5) = 10$. Formula: $c/a = 10/1 = 10$. (Verified)

❌ Common Mistakes

> [!WARNING]
>
> 1. **Sign Errors:** Students often forget the negative sign in the sum of zeroes formula ($-\frac{b}{a}$). Always double-check!

2. **Standard Form:** Before finding $a, b,$ and c, ensure the polynomial is written in standard form ($ax^2 + bx + c$). For example, $5x - x^2 + 3$ must be rewritten as $-x^2 + 5x + 3$.
3. **Degree Check:** An expression like $x^{-2} + 2x + 1$ is **NOT** a polynomial because the exponent is negative.

📝 Quick Revision

Concept	Formula / Property
Quadratic Formula Form	$ax^2 + bx + c = 0$
Sum of Zeros ($\alpha + \beta$)	$-b/a$
Product of Zeros ($\alpha\beta$)	c/a
Max Roots	Equal to the degree of the polynomial.

Summary

Polynomials are fundamental algebraic structures. Understanding how the roots of a polynomial relate strictly to its coefficients provides a powerful tool for solving complex algebraic equations without graphing. Always identify the degree first, structure the polynomial into standard form, and then apply your coefficient formulas!